# Diagrama de clases

Shapes

**<<Abstract>>**
**GeometricShape**

- volume: double
- surface: double

+ getVolume(): double
+ getSurface(): double
+ setVolume(volume: double): void
+ setSurface(surface: double): void

**Cilinder**

- radio: double
- height: double

<<Constructor>>Cilinder(radio:double, height: double)
- calculateVolume(): double
- calculateSurface(): double

**Prism**

- side: double
- height: double
- apothem: double

<<Constructor>> Prism(side: double, height: double, apothem: double)
- calculateBaseArea(): double
- calculateVolume(): double
- calculateSurface(): double

**Piramid**

- base: double
- height: double
- apothem: double

<<Constructor>>Piramid(base: double, height: double, apothem: double)
- calculateVolume(): double
- calculateSurface(): double

**Sphere**

- radio: double

<<Constructor>>Sphere(radio: double)
- calculateVolume(): double
- calculateSurface(): double

**Cube**

- side: double

<<Constructor>>Cube(side: double)
- calculateVolume(): double
- calculateSurface(): double

**ImgRezise**

*+ resize(ImageIcon icon)*: void

**CilinderWindow**

- container: Container
- btnCalculate: JButton
- lblVolume: JLabel
- lblSurface: JLabel
- lblRadio: JLabel
- lblHeight: JLabel
- txtRadio: JTextField
- txtRadio: JTextField

<<constructor>>CilinderWindow()
- start(): void
- ActionPerformed(e: ActionEvent): void

**PrismWindow**

- container: Container
- btnCalculate: JButton
- lblVolume: JLabel
- lblSurface: JLabel
- lblSide: JLabel
- lblHeight: JLabel
- lblApothem: JLabel
- txtSide : JTextField
- txtHeight: JTextField
- txtApothem: JTextField

<<constructor>>PrismWindow()
- start(): void
- ActionPerformed(e: ActionEvent): void

**PiramidWindow**

- container: Container
- btnCalculate: JButton
- lblVolume: JLabel
- lblSurface: JLabel
- lblBase: JLabel
- lblHeight: JLabel
- lblApothem: JLabel
- txtBase: JTextField
- txtHeight: JTextField
- txtApothem: JTextField

<<constructor>>PiramidWindow()
- start(): void
- ActionPerformed(e: ActionEvent): void

**SphereWindow**

- container: Container
- btnCalculate: JButton
- lblVolume: JLabel
- lblSurface: JLabel
- lblRadio: JLabel
- txtRadio: JTextField

<<constructor>>SphereWindow()
- start(): void
- ActionPerformed(e: ActionEvent): void

**CubeWindow**

- container: Container
- btnCalculate: JButton
- lblVolume: JLabel
- lblSurface: JLabel
- lblSide: JLabel
- txtSide: JTextField

<<constructor>>CubeWindow()
- start(): void
- ActionPerformed(e: ActionEvent): void

**MainWindow**

- container: Container
- btnCilinder: JButton
- btnSphere: JButton
- btnPiramid: JButton
- btnCube: JButton
- btnPrism: JButton

<<constructor>>MainWindow()
- start(): void
- ActionPerformed(e: ActionEvent): void

**Shapes**

+ Main(): void